

BioCoLoop: COLLABORATIVE AGENTIC RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

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ABSTRACT

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access. A key challenge for self-improving AI in this setting is to use such distributed evidence to improve both model parameters and the computational solutions built around them. Collaborative learning typically optimizes fixed models, whereas automated model design and research generally rely on centralized training and evaluation signals. Bridging the two is nontrivial: distributed evidence must support not only optimization, but also the evaluation and refinement of computational changes that generalize beyond their local settings. To tackle this, we introduce BioCoLoop, a collaborative research framework in which laboratory-local measurements support parameter fitting and an evidence-guided proposal–evaluation loop. Local training shards update a shared predictor, while local development shards produce aggregate evidence cards that guide the next executable proposal. Across three biological settings spanning drug–target interaction prediction, observed-response proteomics, and cell perturbation, we evaluate BioCoLoop with ten simulated laboratories against single-laboratory fixed-model and direct-optimization baselines. The complete system achieves the highest score on four of five evaluated endpoints. BioCoLoop offers a practical path toward self-improving biological AI under data-sharing constraints.

1. INTRODUCTION

Artificial intelligence is becoming an experimental tool for biology: models predict molecular interactions, connect protein responses to drug effects, and estimate how cells respond to perturbations. The emerging vision of an AI virtual cell places these capabilities within a broader process of generating and testing biological hypotheses (Bunne et al., 2024; Sun et al., 2026). Research agents extend this direction from prediction to model development. They propose computational changes, run experiments and use the results to revise their next proposal (Jiang et al., 2025b; Lu et al., 2024; Zhou et al., 2026). This iterative process offers a route from manually designed predictors toward self-improving biological AI.

A biological research agent, however, usually works with the data available in its own execution environment. Useful evidence may be held elsewhere: one laboratory studies a different cell type, another measures responses under a different stimulus, and a third uses a different experimental assay. Data-sharing restrictions can prevent these measurements from being brought together (Sheller et al., 2020; Bakhtiari et al., 2025). The opportunity is therefore broader than learning from additional examples. Can measurements from different laboratories also inform which model designs to try, which training choices to retain, and what to investigate next, without transferring the underlying data?

Figure 1 contrasts the relevant research settings. Centralized agents use accessible experiment outcomes to revise a predictor. Collaborative training allows several data holders to fit a specified predictor through local computation and model-update aggregation (McMahan et al., 2017; Wang et al., 2024). Our setting makes distributed measurements available for a further purpose: evaluating candidate designs and informing subsequent research decisions. Here, a laboratory denotes a data holder; its computational worker carries out local training and evaluation.

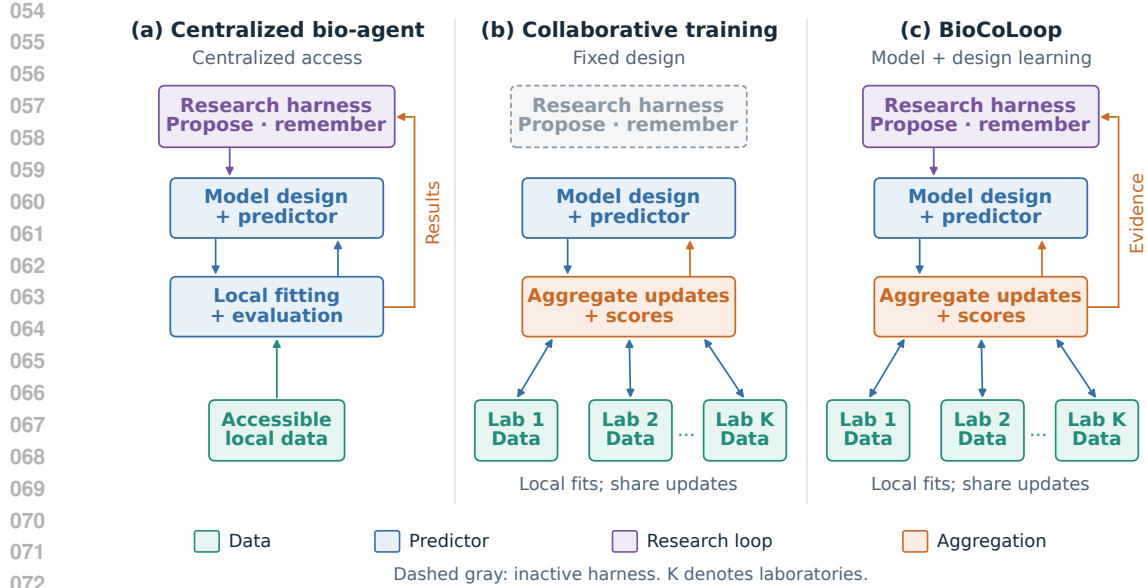


Figure 1: Research settings for biological AI. A centralized bio-agent revises a predictor using accessible data. Collaborative training fits a specified predictor using data held by K laboratories. BioCoLoop makes laboratory-local evaluation evidence available to an iterative research process, so that distributed measurements inform both parameter fitting and subsequent design choices. Colors identify data, predictors, research components and aggregation interfaces.

We introduce BioCoLoop, a collaborative research framework in which laboratory-local evidence drives both predictive learning and iterative model development. A research controller proposes a change to the prediction components or training configuration. Laboratory workers fit and evaluate the resulting candidate, and the coordinator summarizes their measurements in an *evidence card*. Each card records what changed, how training progressed and whether the candidate improved on the current predictor. The history of these cards informs the next proposal. This common research interface supports different biological predictors, while their task-specific inputs and objectives remain with the local training workers.

Our contributions are:

- **A collaborative setting for biological model development.** BioCoLoop provides a common training and research interface through which data holders contribute local model updates and evaluation evidence to the improvement of a task-specific predictor.
- **An evidence-guided research harness.** The loop turns a proposed hypothesis into an executable design, records its measured outcome and uses the accumulated history to guide the next proposal. A separate trajectory-guided allocation study examines how the same measurements can help decide which candidates receive further training.
- **An empirical account of when collaboration helps.** Across drug–target interaction prediction, observed-response proteomics and cell perturbation identification, the complete system leads four of five main endpoints. Controlled studies distinguish gains from additional data, differences among source scenarios, proposal feedback and training allocation. They show that source compatibility and task-specific learning trajectories matter alongside the number of participating laboratories.

2. RELATED WORK

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed

batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2025). BioCoLoop connects parameter fitting to a second laboratory interface for evaluating proposed designs. Aggregate development evidence supports both proposal revision and allocation of further training.

AI for AI in biological research. Research agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist connects ideas to executed experiments, replanning and writing (Lu et al., 2024). DrugEvolve applies research, implementation, evaluation and algorithm memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These systems motivate executable revision and persistent experiment history. Our study asks how such feedback can be supplied by multiple data-owning laboratories through a shared research interface. We instantiate that interface with TAPB for DTI, a ProteinTalks-derived efficacy head for proteomic responses, and scDEBART-based cell representations (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). These models serve as task-specific prediction components; the proposal schema, local training service and evidence-driven selection mechanism remain common across tasks. The strong performance of simple references on some perturbation benchmarks further motivates separating the effect of a research loop from the choice of predictor (Ahlmann-Eltze et al., 2025).

3. METHOD: COLLABORATIVE BIOLOGICAL MODEL IMPROVEMENT

3.1 PROBLEM AND RESEARCH SETTING

Consider K laboratories working on the same biological prediction task. Laboratory i holds training data D_i^{train} , with $n_i = |D_i^{\text{train}}|$. Designated evaluation workers retain local development panels D_j^{dev} . A task specification $\mathcal{T}$ defines the prediction inputs, outputs, training loss and evaluation metric. We write the task predictor as $f_{a,w}$, where $a \in \mathcal{A}$ specifies its design and w its trainable parameters. Given an initial design, a design library $\mathcal{A}$ and a research budget, BioCoLoop seeks a design and fitted predictor that perform well on the designated evaluation population. The three task families share this research procedure while using task-specific predictors.

Our design library varies four choices: learning rate, weight decay, server momentum and an optional residual prediction module. A `design_id` names one configuration of these choices. A research controller selects configurations to investigate, laboratory workers train and evaluate them, and a coordinator returns a concise record of each outcome to the controller (Figure 2). The resulting research history informs subsequent proposals. Raw measurements stay with the workers; model states support parameter learning, while aggregate development measurements support design decisions. The experiments instantiate these roles as separate data-holding workers on one host.

3.2 LOCAL FITTING AND SHARED EVALUATION

For each candidate a , workers initialize identically as $w_0 = \text{Init}(a, z)$, where z is the study seed. Candidates start afresh rather than continuing the incumbent’s training. Let LocalEpoch_a denote one full local AdamW epoch with the candidate’s learning rate and weight decay, batch size 64 and gradient-norm clipping at one. The optimizer state resets at each round. With sample weights $\pi_i = n_i / \sum_{j=1}^K n_j$, server momentum β_a and initial velocity $v_0 = 0$, round r updates the trainable parameters by

$$\begin{aligned} u_{i,r} &= \text{LocalEpoch}_a(w_{r-1}; D_i^{\text{train}}), & \bar{w}_r &= \sum_{i=1}^K \pi_i u_{i,r}, \\ v_r &= \beta_a v_{r-1} + \bar{w}_r - w_{r-1}, & w_r &= w_{r-1} + v_r. \end{aligned} \quad (1)$$

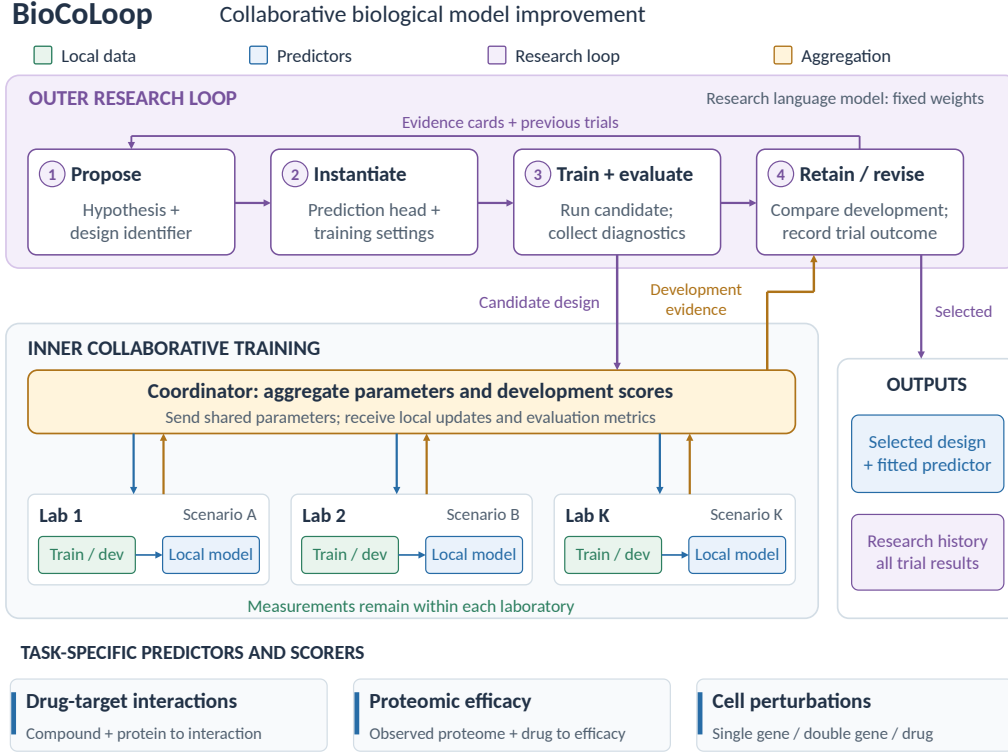


Figure 2: Data flow in BioCoLoop. The outer research loop proposes a model design and receives aggregate development evidence after its evaluation. The inner training process distributes model parameters to K laboratories, fits them locally and aggregates the returned updates. The next proposal uses the accumulated research history. Task adapters implement drug–target interaction prediction, observed-response proteomic efficacy prediction and cell perturbation identification.

Thus $\beta_a = 0$ gives sample-weighted parameter averaging (McMahan et al., 2017). Frozen feature extractors remain unchanged; model-buffer handling is specified in Appendix A. Task adapters supply the biological inputs, trainable prediction components and local loss.

Every five rounds and at the final round, workers evaluate the aggregated model on a protocol-defined set of development panels $\mathcal{E}$. In the main comparison these are the participating laboratories’ panels; fixed-pool repartitioning retains all ten original panels, and source-transfer studies use the original target panels. Define $s_i(a, r)$ and $\ell_i^{\text{dev}}(a, r)$ as the primary score and loss on panel i . The coordinator computes

$$S(a, r) = \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} s_i(a, r), \quad L(a, r) = \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \ell_i^{\text{dev}}(a, r). \quad (2)$$

The selected round $\hat{r}_a$ maximizes $S(a, r)$, with lower $L(a, r)$ breaking ties and the earlier round retained on a complete tie. Its parameters are $\hat{w}_a = w_{\hat{r}_a}$; the pair $q(a) = (S(a, \hat{r}_a), -L(a, \hat{r}_a))$ ranks the design. Training therefore weights examples, whereas selection weights the designated development panels equally. The main proposal-search experiments assign each candidate 100 training rounds; training duration is specified by the evaluation schedule, not by `design_id`.

For auxiliary-source studies with different prediction targets, shared encoders follow the same update rule, while each source-specific head is aggregated only over its owning workers. Checkpoint selection then uses the original target’s development panels. This allows auxiliary measurements to improve the shared representation while preserving each source’s prediction objective.

3.3 PROPOSAL REVISION FROM RESEARCH HISTORY

The first proposal is generated once without development feedback and shared by direct and loop search. History feedback begins with the second proposal. For subsequent research steps t , let a_t be the retained design, H_t the ordered history of previous trials and $\mathcal{U}_t$ the untried configurations. The controller, with fixed language-model weights θ , generates a proposal p_t and the coordinator maps its `design_id`, denoted $\text{id}(p_t)$, to the candidate:

$$p_t \sim P_\theta(\cdot \mid \mathcal{T}, a_t, H_t, \mathcal{U}_t), \quad \tilde{a}_t = \text{Compile}(\text{id}(p_t)). \quad (3)$$

The proposal also contains a `hypothesis`, an `experiment` and an `expected_effect`: these describe the motivation, intended comparison and anticipated outcome. They provide an interpretable account of the proposed experiment. The identifier determines the executed configuration, and measured development performance determines selection.

After training $\tilde{a}_t$, the coordinator creates an *evidence card* e_t : its executed configuration and hypothesis, selected score and loss, selected round, first/last training and development diagnostics, and acceptance decision. Using lexicographic comparison of q —higher score, then lower loss—the research state becomes

$$a_{t+1} = \begin{cases} \tilde{a}_t, & q(\tilde{a}_t) >_{\text{lex}} q(a_t), \\ a_t, & \text{otherwise,} \end{cases} \quad H_{t+1} = H_t \parallel [e_t], \quad (4)$$

where $\parallel$ appends a card to the history. Unsuccessful trials remain in that history; invalid proposals and numerically failed fits consume a proposal slot and leave the incumbent unchanged. Subsequent proposals therefore respond to measured outcomes through updated context, without fine-tuning the research language model. After the proposal budget is exhausted, the retained design and its selected checkpoint are evaluated on the held-out task.

Appendix A gives the executable schema, an example history and the complete algorithm. A separate experiment uses short training trajectories to allocate a fixed training budget among a predetermined set of designs. Its scheduler chooses which candidates receive longer training; it does not generate language-model proposals. This control isolates training allocation from the history-guided proposal process above.

4. EXPERIMENTS

4.1 TASKS, DATASETS AND EVALUATION

We study three biological prediction problems, using a separate predictor for each task and the same BioCoLoop research interface. The main comparison uses ten simulated laboratories formed by disjoint groups within each dataset. A complementary source-as-laboratory study assigns different biological collections or scenarios to separate workers. The first setting controls access to data; the second tests transfer between data sources.

Drug–target interactions. Given a compound and a protein sequence, the predictor estimates whether they interact. BindingDB supplies 50,149 training and 5,604 development pairs; canonical compounds define the client groups. The reference is TAPB with frozen ESM2 protein features and trainable drug, fusion and classification modules (Lin et al., 2025). The main endpoint is AUROC on 5,505 random-test pairs. Tests with 3,791 unseen-drug and 2,507 unseen-protein pairs assess entity generalization. AP, MCC, accuracy and cross-entropy provide secondary measures.

Observed-response proteomics. Given a drug representation and its measured 6h/24h proteomic response, the predictor estimates binary efficacy. A ProteinTalks PTPC subset supplies 386 training, 105 development and 148 test conditions (Sun et al., 2026). Compound-group assignment gives each client 49–50 training-plus-development conditions. The reference is a ProteinTalks-derived efficacy head with protein and drug branches, fitted from random initialization. Average precision (AP) is primary; AUROC, accuracy and cross-entropy are secondary. This task uses observed post-treatment measurements, rather than predicting efficacy from an unmeasured proteome.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
<i>Fixed task models, one laboratory</i>					
TAPB	82.83 $\pm$ 1.22	–	–	–	–
ProteinTalks-derived head	–	<u>25.05</u> $\pm$ 4.56	–	–	–
scDEBART response head	–	–	<u>29.94</u> $\pm$ 0.00	25.56 $\pm$ 0.00	<u>19.40</u> $\pm$ 0.00
Qwen direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59

Table 1: Main held-out comparison across three task families. Each column uses its named task model in all methods. Direct optimization proposes configurations without evaluation history; BioCoLoop adds collaborative access and evidence feedback. Values are mean $\pm$ sample SD over seeds 42–44, multiplied by 100; SD is not a confidence interval. Bold and underline mark the best and second-best displayed means, including ties. A dash indicates a model for a different task. Full factorial results and secondary metrics are in Appendix A.

Cell perturbation identification. Given an expression response and five candidate interventions, the model identifies the intervention whose predicted response has the highest cosine agreement with the observation. We use VCC single-gene perturbations (Arc Institute, 2025), Norman double-gene perturbations (Norman et al., 2019), and a Tahoe subset of drug perturbations in A549 at 5 μ M (Zhang et al., 2025). Intervention-disjoint training/development/test counts are 60/20/20, 70/20/27 and 180/60/67, respectively. The reference is a frozen scDEBART backbone with a trainable response head (Sung & Kim, 2026). All arms use the same corrected attention-mask implementation and fixed candidate rosters. Primary scoring is intervention-macro Top-1: accuracy is first averaged within each intervention, then across interventions. Random choice has expectation 20%. Micro Top-1, MRR, Top-3 and cross-entropy are secondary. Appendix A describes response construction, chemical conditioning, losses and secondary metrics.

4.2 COMPARED METHODS AND SHARED PROTOCOL

The *fixed model* trains each reference architecture with a fixed recipe on preselected laboratory 0. *Direct optimization* uses Qwen2.5-7B-Instruct to propose designs without evaluation history, using the same laboratory. *BioCoLoop* supplies evidence-card feedback to the same proposer and trains across ten laboratories. Thus the main comparison measures the benefit of the complete collaborative research setting. A six-arm factorial comparison crosses fixed/direct/loop research with one/ten laboratories to separate data access from proposal feedback.

Direct and loop search share a twelve-design library, six proposal slots and the first proposal. Every candidate starts from the same task/seed initialization and receives 100 training rounds, with development evaluation every five rounds. Selection uses equal-weight development scores over participating laboratories. The main results report mean and sample SD over seeds 42–44 on fixed data partitions. All methods within a task share the held-out test set and scorer. Appendix A gives the full factorial comparison, complete optimization settings and uncertainty estimates.

4.3 MAIN PERFORMANCE AND ATTRIBUTION

Table 1 displays the three task families together. BioCoLoop leads four of five endpoints: DTI AUROC reaches 93.76%, proteomic AP reaches 36.99%, and VCC and Tahoe Top-1 reach 31.54% and 22.39%. Compared with one-laboratory direct optimization, the three cell gains are +10.45, +8.10 and +6.97 percentage points. The one-laboratory fixed model remains best on Norman at 25.56%, versus 22.25% for BioCoLoop.

What drives these differences? Holding loop search fixed, increasing participation from one to ten laboratories adds +3.87 AUROC points for DTI, +16.85 AP points for proteomics, and +11.32, +8.10 and +6.97 Top-1 points for the cell tasks. At fixed access, direct and loop select identical checkpoints in this three-seed, six-proposal experiment. Their identical scores are therefore outcomes of the com-

Table 2: Fixed-design source–scenario ablation on VCC. K : training clients; N : training intervention conditions. The two $K = 4$, $N = 60$ arms share a 15-condition VCC anchor and differ in the sources of the other three clients. All arms use 100 rounds and the same target evaluation. Mean $\pm$ sample SD over three seeds; best means are bold. Cell counts and gene panels remain source-specific; protocol details are in Appendix B.

Training sources	K	N	Top-1 (%) $\uparrow$	MRR $\uparrow$
VCC, limited data	1	15	29.36 ± 0.26	0.5697 ± 0.0019
VCC, all target data	1	60	29.43 ± 0.09	0.5600 ± 0.0009
VCC, same-source split	4	60	30.70 ± 0.42	0.5624 ± 0.0004
VCC + external studies	4	60	28.82 ± 0.00	0.5630 ± 0.0000

pleted searches. The following studies examine source composition, search trajectories and training allocation separately.

4.4 LEARNING FROM DIFFERENT LABORATORY SCENARIOS

In source-as-laboratory experiments, each auxiliary collection remains a separate training client. The target’s original development panels select checkpoints, and a common target test measures transfer. For DTI, BindingDB, BioSNAP, Davis and Human form four collection-defined clients. For VCC, the auxiliary clients are Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562: they differ in study, cellular background or stimulation, rather than being random shards of VCC. Appendix B gives their identities, sizes and exclusions.

DTI target-only AUROC is 85.47; Davis and Human raise it to 87.86 and 87.95, whereas BioSNAP and all three combined yield 82.44 and 84.30. Adding Jiang raises VCC Top-1 from 29.49 to 33.07; combining all sources gives 27.33. The Jiang change is +3.58 points, with paired intervention-bootstrap 95% CI $[-2.11, 9.51]$.

For proteomics, we assign each of four studies to one training laboratory: the target PTPC collection, Lin, Ruprecht and decryptE (Lin et al., 2026; Ruprecht et al., 2020; Eckert et al., 2025). A source-aware adapter shares encoders and retains private response heads to accommodate their different supervised objectives. Under the same fixed recipe and target-only development selection, adding all three external studies raises target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Table B.7 reports all four controls; Appendix B also retains the separate protocol with ten target training shards. Together, these studies show that external measurements can improve target prediction, while source compatibility matters alongside data volume.

To compare source composition at matched participation and condition count, we fix a 15-intervention VCC anchor client and add three clients of 15 interventions each: either further VCC conditions or one collection from each external study. Table 2 shows that cross-scenario training reaches 28.82% Top-1, compared with 30.70% for the same-source split; MRR is nearly unchanged (0.5630 versus 0.5624). Under this fixed design, heterogeneous updates do not improve target identification. Matching K and intervention counts isolates this comparison from increasing client count or condition count, while source-specific cell coverage and gene panels remain part of the distribution shift.

4.5 HOW MANY LABORATORIES SHOULD PARTICIPATE?

We vary $K \in \{1, 2, 5, 10\}$ in two ways: adding the first K clients increases available data; merging the full pool into K groups holds total data fixed. Both use the default design and 100 rounds. The fixed-pool comparison retains the original ten development panels. DTI uses seed 61; the other endpoints use seeds 61–63.

Figure 3 separates the two effects. Increasing participation raises DTI AUROC from 83.65 to 93.88. With the DTI pool fixed, $K = 1, 2, 5, 10$ gives 85.47, 88.96, 88.05 and 93.88. The fixed-pool optimum varies by task: proteomics reaches 39.19 AP at $K = 5$, whereas Norman falls from 49.63 to 15.48 Top-1 between $K = 1$ and $K = 10$. Smaller local partitions change optimizer steps and the

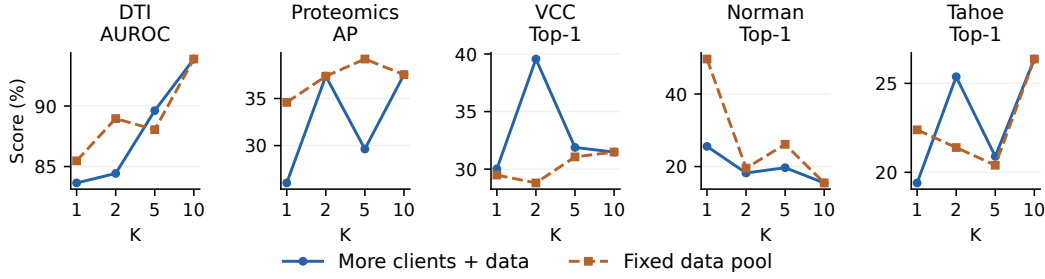


Figure 3: Laboratory-count sensitivity. Solid lines add participating clients and data; dashed lines repartition the full training pool. Points are means over the study seeds; full values and SD are in Appendix B. Each panel uses its task’s primary metric; tested K settings are equally spaced.

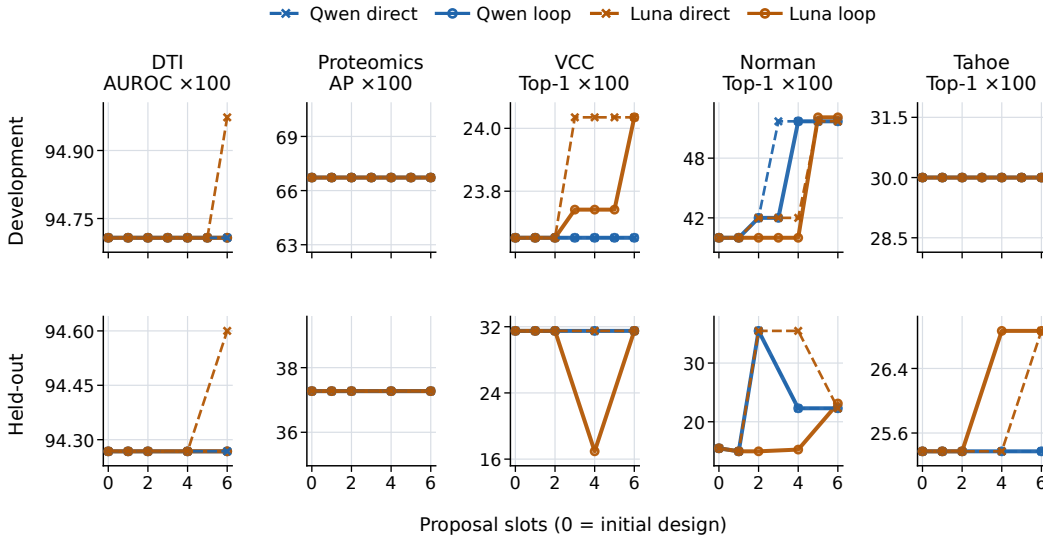


Figure 4: Research trajectories with and without evidence history. Upper panels show the development selection score; lower panels show held-out scores of the development-selected checkpoint at each prefix. Each candidate receives 100 rounds. Slot 0 is the initial design. The two research models use the same design library within each task.

cell objective’s contrastive negative pool. More participants therefore do not automatically imply a better optimization setting.

4.6 HOW DOES FEEDBACK CHANGE THE RESEARCH TRAJECTORY?

We compare Qwen2.5-7B-Instruct and the gpt-5.6-luna service under a common six-slot protocol on all five endpoints. Direct and loop share the twelve designs, initial proposal, training schedule and selection rule; only loop receives measured history. DTI uses seed 42 and the other tasks seed 61. Figure 4 shows development-selected test performance at prespecified proposal prefixes.

Across ten direct/loop pairs, Norman/Luna improves by 0.81 Top-1 points, DTI/Luna decreases by 0.33 AUROC points, and eight pairs tie. A separate 36-design, 24-slot study on proteomics and the three cell tasks reveals differences in search speed. Luna reaches Norman’s common development target at slot 4 with feedback versus slot 22 without it; both finish at 23.11% test Top-1. Direct reaches Tahoe’s common target sooner (5 versus 13 slots), and proteomics retains its initial design. The eight terminal comparisons give seven ties and one decrease with feedback. Qwen’s VCC test Top-1 falls from 31.50 at prefix 6 to 19.61 at prefix 24 despite improved development selection. Thus the useful

Benchmark	Uniform	Evidence-guided	Gain	W/T/L
Proteomics: efficacy	37.47	37.47	+0.00	0/3/0
Cell: VCC single-gene	20.01	20.01	+0.00	0/3/0
Cell: Norman double-gene	11.48	22.32	+10.84	3/0/0
Cell: Tahoe drug	20.90	22.89	+1.99	1/2/0

Table 3: Does early evidence improve training allocation? Held-out scores use the same ten designs, ten laboratories, 800 full-client rounds and 160 aggregate development evaluations per method. Uniform allocation gives every design 80 rounds; the evidence-guided scheduler screens all designs for 20 rounds and restarts six promoted designs for 100-round training. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [-1.00, 7.46]. The separate DTI development replay is in Appendix A.

research budget depends on the task: faster development progress and better final generalization are distinct outcomes. Appendix B connects these trajectories to the executed configuration changes.

4.7 CAN MEASURED TRAJECTORIES IMPROVE TRAINING ALLOCATION?

To isolate allocation from proposal generation, we give both schedulers the same ten designs and all ten laboratories. Uniform allocation trains every design for 80 rounds. Evidence-guided allocation screens every design for 20 rounds, then restarts six promoted designs for 100-round training. Both spend 800 full-client rounds and 160 aggregate development evaluations. The promotion rule combines the best screening score with improvement from round 5 to round 20 (Appendix A).

With proposal generation held fixed, evidence-guided allocation yields four wins and eight ties across twelve task-seed comparisons (seeds 53–55). Norman improves from 11.48% to 22.32% Top-1, a +10.84-point mean effect with hierarchical paired-bootstrap 95% CI [4.54, 17.73]. A promoted low-learning-rate residual design benefits from training beyond uniform allocation’s stopping point. Tahoe gains 1.99 points; VCC and proteomics tie. This result supports using learning trajectories to decide where to spend training compute, separately from the proposal-history effects in Section 4.6.

5. CONCLUSION

BioCoLoop connects laboratory-local parameter fitting to an evidence-guided research loop through shared proposal, training and evaluation interfaces. The complete system leads four of five main-comparison endpoints, and compute-matched allocation improves Norman Top-1 by 10.84 points on average. Source-defined laboratory studies show transfer gains in DTI and proteomics, while cell comparisons highlight the importance of source compatibility. Laboratory-count and dual-proposer studies reveal how partitioning and research budget shape learning. The results distinguish parameter-learning gains, efficient allocation and proposal-history effects, providing an empirical basis for collaborative biological model development and more effective research feedback.

REPRODUCIBILITY STATEMENT

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, data partitions, reference models and metrics. Appendix A provides the executable proposal schema, complete design library, optimization settings and matched-budget promotion rule. Appendix B specifies the laboratory-count, source-transfer and proposer-budget studies. We will release the implementation, task adapters, experiment configurations and prediction-level evaluation records. The reproducibility artifact records code and data hashes, random seeds, proposals, development trajectories, selection decisions and saved predictions. Aggregate tables are generated from completed evaluation records, with independent score recomputation for the main comparison and completed follow-up evaluations. The machine-readable supplement includes all 504 attempted proposal slots and their configuration changes; the appendix presents representative decisions and complete performance summaries. These records connect each reported result to its configuration and selected checkpoint.

ETHICS STATEMENT

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Laboratory boundaries are simulated by local workers. Retaining raw data at workers avoids direct transfer to the research model; parameter updates and aggregate diagnostics would still require an explicit privacy and security policy in a multi-institutional deployment. Predictions are evaluated as research outputs, not as validated biological or therapeutic interventions.

AI USE STATEMENT

Local Qwen2.5-7B-Instruct generates proposals in the main study; the proposer sensitivity study additionally uses the service identified as gpt-5.6-luna. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, manuscript drafting and revision, and the Chinese reading companion. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

REFERENCES

- Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8):1657–1661, 2025. doi: 10.1038/s41592-025-02772-6. URL <https://doi.org/10.1038/s41592-025-02772-6>.
- Arc Institute. Virtual cell challenge. Arc Virtual Cell Atlas dataset, 2025. URL <https://github.com/ArcInstitute/arc-virtual-cell-atlas/tree/4f7bb170c61d2fafce70f67db6e52c7d9af1be4b/virtual-cell-challenge>. 2025 data release; training file adata_Training.h5ad. Documentation accessed September 22, 2026.
- Mohammad Bakhtiari, Stefan Bonn, Fabian Theis, Olga Zolotareva, and Jan Baumbach. FedscGen: privacy-preserving federated batch effect correction of single-cell RNA sequencing data. *Genome Biology*, 26:216, 2025. doi: 10.1186/s13059-025-03684-6. URL <https://doi.org/10.1186/s13059-025-03684-6>.
- Charlotte Bunne, Yusuf Roohani, Yanay Rosen, Ankit Gupta, Xikun Zhang, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell*, 187(25):7045–7063, 2024. doi: 10.1016/j.cell.2024.11.015. URL <https://doi.org/10.1016/j.cell.2024.11.015>. Preprint version 2.
- Jiayuan Ding, Jianhui Lin, Ziyang Miao, Nils Mechtel, Shiyu Jiang, Yixin Wang, Zhaoyu Fang, Jorge D. Martin-Rufino, Chen Weng, Reuben Saunders, Weize Xu, Jonathan S. Weissman, Min Li, Jiliang Tang, Wei Ouyang, Yuancheng Ryan Lu, and Xiaojie Qiu. Predictive single cell foundation model for gene regulation and aging with privacy-preserving tabular learning, 2026. URL <https://arxiv.org/abs/2607.19400v1>. Version 1.
- Stephan Eckert, Nicola Berner, Karl Kramer, Annika Schneider, Julian Müller, Severin Lechner, Sarah Brajkovic, Amirhossein Sakhteman, Christian Graetz, Jonas Fackler, Michael Dudek, Michael W. Pfaffl, Percy Knolle, Stephanie Wilhelm, and Bernhard Kuster. Decrypting the molecular basis of cellular drug phenotypes by dose-resolved expression proteomics. *Nature Biotechnology*, 43:406–415, 2025. doi: 10.1038/s41587-024-02218-y. URL <https://doi.org/10.1038/s41587-024-02218-y>. First published online 7 May 2024.
- GAIR-NLP DrugEvolve contributors. DrugEvolve: Official source and applications, 2026. URL <https://github.com/GAIR-NLP/DrugEvolve/tree/e945c7d0da28ddce73904419116a51e03921e6e6>. Commit e945c7d0da28ddce73904419116a51e03921e6e6.
- Longda Jiang, Carol Dalgarno, Efthymia Papalexi, Isabella Mascio, Hans-Hermann Wessels, Huiyong Yun, Nika Iremadze, Gila Lithwick-Yanai, Doron Lipson, and Rahul Satija. Systematic

- reconstruction of molecular pathway signatures using scalable single-cell perturbation screens. *Nature Cell Biology*, 27:505–517, 2025a. doi: 10.1038/s41556-025-01622-z. URL <https://doi.org/10.1038/s41556-025-01622-z>.
- Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. AIDE: AI-driven exploration in the space of code, 2025b. URL <https://arxiv.org/abs/2502.13138v1>. Version 1, 18 February 2025.
- Mikhail Khodak, Renbo Tu, Tian Li, Liam Li, Maria-Florina Balcan, Virginia Smith, and Ameet Talwalkar. Federated hyperparameter tuning: Challenges, baselines, and connections to weight-sharing. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://papers.neurips.cc/paper_files/paper/2021/hash/a0205b87490c847182672e8d371e9948-Abstract.html.
- Haoyuan Li, Mathias Funk, and Aaqib Saeed. Helmsman: Autonomous synthesis of federated learning systems via collaborative LLM agents, 2025. URL <https://arxiv.org/abs/2510.14512v2>. Version 2, 19 December 2025.
- Chuwei Lin, Catherine M. Sniezek, Christopher D. McGann, Rashmi Karki, Ross M. Giglio, Benjamin A. Garcia, José L. McFaline-Figueroa, and Devin K. Schweppe. Defining the heterogeneous molecular landscape of lung cancer cell responses to epigenetic inhibition. *Communications Biology*, 9:176, 2026. doi: 10.1038/s42003-025-09455-0. URL <https://doi.org/10.1038/s42003-025-09455-0>. Published 28 January 2026.
- Gaoming Lin, Xin Zhang, Zhonghao Ren, Quan Zou, Prayag Tiwari, Changjun Zhou, and Yijie Ding. TAPB: an interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. *Nature Communications*, 16:10867, 2025. doi: 10.1038/s41467-025-66915-1. URL <https://doi.org/10.1038/s41467-025-66915-1>.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery, 2024. URL <https://arxiv.org/abs/2408.06292v1>. Version 1, 12 August 2024.
- Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pp. 1273–1282, 2017. URL <https://proceedings.mlr.press/v54/mcmahan17a.html>.
- Nils Mechtel, Wei Ouyang, and Chiron Platform contributors. Chiron Platform: Federated single-cell training and agent interfaces, 2026. URL <https://github.com/aristoteleo/chiron/tree/3241b35c8b516c126aa68da450708de3da717ee0>. Commit 3241b35c8b516c126aa68da450708de3da717ee0.
- Ajay Nadig, Joseph M. Replogle, Angela N. Pogson, Mukundh Murthy, Steven A. McCarroll, Jonathan S. Weissman, Elise B. Robinson, and Luke J. O’Connor. Transcriptome-wide analysis of differential expression in perturbation atlases. *Nature Genetics*, 57:1228–1237, 2025. doi: 10.1038/s41588-025-02169-3. URL <https://doi.org/10.1038/s41588-025-02169-3>.
- Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019. doi: 10.1126/science.aax4438. URL <https://doi.org/10.1126/science.aax4438>.
- Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, Jeffrey A. Hussmann, Alexander Lenail, Alina Guna, Lauren Mascibroda, Eric J. Wagner, Karen Adelman, Gila Lithwick-Yanai, Nika Iremadze, Florian Oberstrass, Doron Lipson, Jessica L. Bonnar, Marco Jost, Thomas M. Norman, and Jonathan S. Weissman. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013. URL <https://doi.org/10.1016/j.cell.2022.05.013>.

- Benjamin Ruprecht, Julie Di Bernardo, Zhao Wang, Xuan Mo, Oleg Ursu, Matthew Christopher, Rafael B. Fernandez, Li Zheng, Brian D. Dill, Huijun Wang, Yuting Xu, Andy Liaw, Jonathan D. Mortison, Nirodhini Siriwardana, Brian Andresen, Meir Glick, James R. Tata, Victoria Kutilek, Ivan Cornella-Taracido, and An Chi. A mass spectrometry-based proteome map of drug action in lung cancer cell lines. *Nature Chemical Biology*, 16:1111–1119, 2020. doi: 10.1038/s41589-020-0572-3. URL <https://doi.org/10.1038/s41589-020-0572-3>.
- Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10:12598, 2020. doi: 10.1038/s41598-020-69250-1. URL <https://doi.org/10.1038/s41598-020-69250-1>.
- Rui Sun, Liujia Qian, Yongge Li, Tong Liu, Honghan Cheng, Xuedong Zhang, Xueya Zhou, Yuecheng Zhan, Guangmei Zhang, Zhengchao Luo, Kunpeng Ma, Chunlong Wu, Dongchen Ji, Zhangzhi Xue, Hongxue Meng, Yuhang Xiang, Dingwei Lei, Qianhe Zhou, Wenbin Hu, Yuhan Deng, Lingling Tan, Qi Xiao, Zhiwei Liu, Lei Zeng, Liqin Qian, Xuan Zheng, Qiong Hu, Nianzi Luo, Weinan E, Peijie Zhou, Han Wen, Yi Zhu, and Tiannan Guo. An operational perturbation proteomics-based virtual cell model. *Nature*, 2026. doi: 10.1038/s41586-026-11001-9. URL <https://doi.org/10.1038/s41586-026-11001-9>. Published online 9 September 2026.
- Jieun Sung and Wankyu Kim. scDEBART: Predicting in silico single-cell perturbation responses via large-scale differential expression learning. In *International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=pJyidZg93y>.
- Shuang Wang, Bochen Shen, Lanting Guo, Mengqi Shang, Jinze Liu, Qi Sun, and Bairong Shen. scFed: federated learning for cell type classification with scRNA-seq. *Briefings in Bioinformatics*, 25(1):bbad507, 2024. doi: 10.1093/bib/bbad507. URL <https://doi.org/10.1093/bib/bbad507>.
- Jesse Zhang, Airol A. Ubas, Richard de Borja, Valentine Svensson, Nicole Thomas, Neha Thakar, Aidan Winters, Umair Khan, Matthew G. Jones, John D. Thompson, Vuong Tran, Joseph Pangallo, Efthymia Papalexi, Ajay Sapre, Hoai Nguyen, Oliver Sanderson, Maria Nigos, Olivia Kaplan, Sarah Schroeder, Bryan Hariadi, Simone Marujo, Crina Curca Alec Salvino, Guillermo Gallareta Olivares, Ryan Koehler, Gary Geiss, Alexander Rosenberg, Charles Roco, Daniele Merico, Nima Alidoust, Hani Goodarzi, and Johnny Yu. Tahoe-100M: A giga-scale single-cell perturbation atlas for context-dependent gene function and cellular modeling. *bioRxiv*, 2025. doi: 10.1101/2025.02.20.639398. URL <https://www.biorxiv.org/content/10.1101/2025.02.20.639398v3>. Version 3, May 10, 2025.
- Zhimeng Zhou, Yang Nan, Minjie Mou, Yuntao Qian, Yixiu Liu, Zhengyu Zuo, Hao Yang, Weixian Xu, Bo Li, Wanghao Jiang, Yanlin Ren, Yang Liao, Yimeng Wang, Yinghong Li, Qingxia Yang, Zhiheng Xi, Tiantian Mi, Huaicheng Sun, Pengfei Liu, and Feng Zhu. An AI system for autonomous algorithm evolution in drug development. *bioRxiv*, 2026. doi: 10.64898/2026.08.16.745117. URL <https://doi.org/10.64898/2026.08.16.745117>. Version 1, posted 20 August 2026.

APPENDIX A. ALGORITHM AND TASK PROTOCOLS

A.1 RESEARCH ALGORITHM AND OPTIMIZATION

All three task families use the parameter update, development-selection rule and proposal interface of Section 3. A task adapter defines the input representation, prediction components, training loss and development scorer. The research process below starts from the task’s reference design. The fitted reference also supplies its first evidence card.

BioCoLoop research procedure	
1	Fit the reference design from the task initialization and seed; retain its best development checkpoint and initialize the research history.
2	Generate the shared first proposal without development feedback. From the second slot onward, supply the task, untried configurations, retained design and prior evidence cards.
3	Parse the proposal and map its <code>design_id</code> to an executable configuration. Allow at most two generation attempts in the slot.
4	Fit a valid candidate from the same initialization procedure and seed for the assigned number of aggregation rounds.
5	Select its best development checkpoint. Replace the incumbent only if its score/loss pair is better.
6	Append the executed configuration, hypothesis, diagnostics and decision to the history, including rejected or failed trials.
7	After the final slot, return the retained design, its selected checkpoint and the trial history; score that checkpoint on the held-out task.

In the controlled direct/loop comparison, the first proposal is generated once without development feedback and shared; measured history enters loop search from the second proposal. Direct search subsequently receives the default configuration and untried identifiers without development outcomes; history-guided search receives the retained configuration and accumulated cards. Both searches use the same design library, proposal count and candidate-training schedule. A configuration evaluated previously under the same task, data access and seed can reuse that completed fit; its logical position in each search remains unchanged.

Each aggregation round contains one local AdamW epoch per participating laboratory. The batch size is 64, gradient-norm clipping is one, and the local optimizer state resets each round. The learning rate and weight decay follow the selected design. The proximal penalty is zero throughout the reported design libraries. The coordinator weights local parameter states by their number of training examples. Server momentum follows Equation 1 and applies only to trainable parameters. Floating-point buffers use sample-weighted averaging; integer and Boolean buffers use an elementwise maximum. Every candidate starts with the same task-specific public or random initialization procedure and study seed. The optional residual module has width 32 and a zero-initialized output, while the original prediction head remains trainable.

Full-training candidates receive 100 aggregation rounds, with development evaluation every five rounds. The selection key is mean primary score followed by negative mean loss over the development panels designated by the protocol (Section 3.2). In the main experiment these panels are the participating laboratories. A complete tie retains the earlier checkpoint or incumbent. Training-example counts refer to the adapter’s training rows: for the cell tasks these are intervention-level training profiles, rather than the number of cells underlying a profile.

A.2 DESIGN LIBRARY, PROPOSALS AND EVIDENCE

The prefix D stands for *design*; its two-digit suffix is an identifier, not a training duration or ranking. D00 denotes the starting recipe. The main study selects six further proposals from D01–D11 without repeating a design within a search. Each identifier maps to the complete four-factor configuration below. The extended library and its identifiers are given in Appendix B.

Design	Learning rate	Weight decay	Momentum	Residual module
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

The main-study controller is Qwen2.5-7B-Instruct, using bfloat16 inference, temperature 0.7, top- p 0.9 and at most 512 new tokens. For zero-indexed proposal slot t and generation attempt $b \in \{0, 1\}$, its sampling seed is $z + t + 10,000b$, where z is the study seed. An invalid identifier, a repeated configuration or a malformed response triggers the second attempt; a second failure consumes the slot without fitting a candidate. A numerically failed fit likewise leaves the incumbent unchanged and is recorded as negative evidence.

A proposal has exactly four fields:

```
{
  "hypothesis": "why the proposed change may help",
  "experiment": "the intended controlled comparison",
  "expected_effect": "the anticipated metric or diagnostic change",
  "design_id": "D08"
}
```

The three descriptive fields must be nonempty strings. The identifier must belong to the available design library. The executed configuration is obtained from that identifier; the text records the controller’s rationale, not an additional executable instruction or a measured result.

An evidence card contains the executed configuration, its hypothesis, the best development score and loss, the selected round, and the acceptance decision. Its trajectory summary includes the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the last-minus-first changes in development score and loss. For a 100-round fit these endpoints are rounds 5 and 100. In the main comparison, the development summary also records the worst laboratory score and the number of evaluating laboratories. Training participation and the designated evaluation panels are specified separately for source-transfer and repartitioning studies. Ordered cards are supplied as context; unsuccessful trials are not removed.

An observed proposal sequence. For Norman with ten laboratories and seed 42, D06 is accepted in the first slot: it ties D00’s development Top-1 of 0.400 but reduces development loss from 1.589766 to 1.589408. In the second slot, D08 retains the residual module and changes the learning rate from 10^{-3} to 10^{-4} . Its development Top-1 rises from 0.102 at round 5 to 0.519 at round 100; its development loss falls from 1.681246 to 1.538505. The candidate is accepted. The next proposed design, D07, reaches a best Top-1 of 0.450 and is rejected. Thus the history records both the slower-improving accepted candidate and the later unsuccessful alternative.

Three representative trajectories are summarized below. Scores use the 0–1 scale and proposal slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	0.665833	0.521161	5 / 5
VCC, seed 43	D00	0.240350	1.607792	initial baseline
Norman, seed 42	D08	0.519000	1.538505	3 / 2

PTPC selects stronger weight decay through the loss tie-break at equal AP, whereas VCC retains its initial design. Across the 15 ten-laboratory task–seed trajectories, the two-proposal history-guided

search is better in two, tied in thirteen and worse in none, with a mean development-primary difference of +0.78 percentage points. At six proposals, direct and history-guided search select the same checkpoints. These observations describe search progress within the twelve-design library.

A.3 INDEPENDENT TRAINING-ALLOCATION CONTROL

This study fixes the candidate set and changes only the allocation of training rounds. It uses seeds 53–55 on proteomics and the three cell settings. The ten configurations cover the following factor combinations; every configuration uses the same ten laboratories.

Learning rate	Weight decay	Momentum	Residual module
0.001	0.001	0	no
0.0001	0.001	0	no
0.0001	0.001	0	yes
0.0001	0.01	0.9	no
0.0001	0.01	0.9	yes
0.001	0.001	0.9	yes
0.01	0.001	0	yes
0.01	0.001	0.9	yes
0.01	0.01	0.9	no
0.001	0.01	0	yes

Uniform allocation trains every design for 80 rounds. Trajectory-guided allocation first trains each for 20 rounds. Within each of the three learning-rate groups, it promotes the design with the highest primary score at its best screening checkpoint, using lower development loss to break ties. From the remaining designs in that group, it also promotes the largest development-score improvement from round 5 to round 20, using lower round-20 development loss to break ties. Complete ties follow the order of the configurations in the table. The six promoted designs restart from their initial parameters and train for 100 rounds. Final selection considers their full-training checkpoints; the four unpromoted designs stop after screening.

Both schedules use 800 aggregation rounds: $10 \times 80 = 10 \times 20 + 6 \times 100$. With ten participating laboratories, this is 8,000 local epochs. Evaluation every five rounds gives 160 aggregate development evaluations, each formed from ten laboratory evaluations. These are matched training and evaluation counts; the scheduler itself generates no language-model proposals.

Across the 12 task–seed held-out comparisons, allocation gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties; VCC and PTPC tie in all three seeds. A separate TAPB development replay uses the seven-design intersection of existing trajectories: uniform allocation receives 525 candidate-rounds and trajectory-guided allocation receives 520, with a mean development AUROC difference of +0.39 points (two wins and one tie). It is not included in the twelve held-out comparisons.

A.4 DATA CONSTRUCTION AND TASK ADAPTERS

Drug–target interaction. The BindingDB partition contains 50,149 fitting and 5,604 development pairs. Canonical compounds are assigned to ten laboratories, giving 4,932–5,058 training and 517–643 development pairs per laboratory. The fixed test sets contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. TAPB uses frozen public protein features and trainable drug, fusion and classification components; an optional residual adapter precedes its native confounder-averaged prediction. Classification NLL and masked-language loss form its local training objective.

Observed-response proteomics. The PTPC source pool contains 333 compound groups and 491 conditions, partitioned into 386 fitting and 105 development conditions. A laboratory owns 35–42 fitting conditions; the test set contains 148 conditions. For each 6h and 24h profile, preprocessing applies log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. The two 5,171-dimensional proteomic blocks and drug vector form a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head has protein and drug convolutional branches followed by a

64–32–1 classifier, totaling 464,193 parameters under D00. It is trained from random initialization using binary cross-entropy. All comparison methods use these same inputs and architecture; AP is the selection metric.

Cell inputs and response units. VCC, Norman and Tahoe have whole-intervention fitting/development/test splits of 60/20/20, 70/20/27 and 180/60/67. Training uses one mean response profile per intervention. VCC development and test queries average disjoint groups of up to 64 cells, omitting a final group smaller than 16 cells. Norman queries are the released scVI-derived differential-expression profiles for each gene pair. Tahoe uses one aggregate response profile per compound at the selected cell context and dose. These are the units scored by the identification task.

The scDEBART backbone receives the shared untreated reference and candidate perturbation encoding; its features are fixed while the response head is trained. The corrected feature extractor implements three attention masks as additive blocking masks. Its correction preserves the pretrained weights, gene panels, response targets, intervention partitions and candidate options. Genetic-task scoring excludes the union of directly perturbed genes. Tahoe concatenates a fixed 128-dimensional projection of a 2,048-bit Morgan fingerprint to the 768-dimensional backbone features, with the additional head-input weights initialized to zero. Its drug projection uses seed 1234. The training loss is response MSE plus 0.1 times a contrastive loss over interventions in the local training minibatch; the cosine-logit temperature is 0.1.

Five-option queries and scoring. For every development or test intervention, four distinct distractors are sampled uniformly without replacement from the *same* development or test intervention pool, excluding the true intervention. The true intervention is added and the five positions are shuffled. A single roster is fixed per intervention and reused for all of its response queries, all training seeds and all methods. Split construction and roster sampling use the preparation generator initialized with seed 20260918. Training does not use these five-option rosters; its contrastive negatives come from the local training minibatch.

For each query, predicted candidate responses and the observed response are L2-normalized over the scoring genes. Candidates are ranked by their cosine similarity; equal scores retain the fixed shuffled option order. Let G be the number of evaluated interventions, m_g the number of response queries for intervention g , and $r_{g,j}$ the rank of the true option for its j th query. The primary score is

$$\text{MacroTop1} = \frac{1}{G} \sum_{g=1}^G \frac{1}{m_g} \sum_{j=1}^{m_g} \mathbf{1}\{r_{g,j} = 1\}.$$

Thus each intervention receives equal weight even when it contributes a different number of query profiles. Micro Top-1, mean reciprocal rank and Top-3 average over individual queries. The development loss is cross-entropy over the five cosine scores divided by temperature 0.1. Development selection first computes these scores within each laboratory and then averages laboratories equally.

A.5 COMPLETE COMPARISONS AND UNCERTAINTY

The following tables give the full one/ten-laboratory by fixed/direct/history-guided comparison, method contrasts, proposal-prefix results, secondary metrics and all three drug–target test sets. Bold marks the best displayed value within a task and metric, including ties. Higher is better except for binary cross-entropy and cell cross-entropy.

Labs	Method	DTI Proteomics		Cell perturbation		
		AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	25.56 ±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Fixed model	91.80±2.55	37.07 ±0.08	31.54 ±0.08	15.48±0.00	25.87 ±1.72
10	Direct	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
10	Our loop	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

Table A.1: Six-arm ablation of the task-reference models. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

Held-out contrast	DTI	Proteomics	VCC	Norman	Tahoe
Loop – fixed (1 lab)	+7.06	-4.91	-9.71	-11.41	-3.98
Loop – fixed (10 labs)	+1.96	-0.08	+0.00	+6.77	-3.48
Loop – direct (1 lab)	+0.00	+0.00	-0.87	+0.00	+0.00
Loop – direct (10 labs)	+0.00	+0.00	+0.00	+0.00	+0.00
10 – 1 labs (both fixed)	+8.97	+12.02	+1.61	-10.07	+6.47
10 – 1 labs (both direct)	+3.87	+16.85	+10.45	+8.10	+6.97
10 – 1 labs (both loop)	+3.87	+16.85	+11.32	+8.10	+6.97
Completed seeds ($n/3$)	3/3	3/3	3/3	3/3	3/3

Table A.2: Complete decomposition into design-search improvement (loop minus fixed), feedback-specific improvement (loop minus matched-budget direct), and laboratory-participation improvement (ten minus one lab with loop fixed). Mean paired held-out differences are in percentage points over the three seeds and are computed before rounding.

Proposal budget	Loop better	Tie	Direct better	Mean Δ primary points
	task-seed trajectories			
1	0	15	0	+0.00
2	2	13	0	+0.78
3	0	14	1	-0.04
4	0	15	0	+0.00
5	0	14	1	-0.02
6	0	15	0	+0.00

Table A.3: Executed-prefix analysis of all 15 ten-laboratory development trajectories at common proposal budgets. At two proposals, feedback-guided search improves TAPB seed 44 by 4.84 points and Norman seed 42 by 6.90 points, with thirteen ties and no regressions.

Table A.4: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	0.8283	0.8157	0.4995	0.5237
Direct	1	0.8989	0.8777	0.6571	0.6102
Our loop	1	0.8989	0.8777	0.6571	0.6102
Fixed model	10	0.9180	0.8936	0.7239	0.3654
Direct	10	0.9376	0.9273	0.7371	0.3659
Our loop	10	0.9376	0.9273	0.7371	0.3659
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	0.2505	0.6549	0.4265	0.8514
Direct	1	0.2013	0.6180	0.4434	0.8176
Our loop	1	0.2013	0.6180	0.4434	0.8176
Fixed model	10	0.3707	0.7406	0.3978	0.8514
Direct	10	0.3699	0.7387	0.3983	0.8514
Our loop	10	0.3699	0.7387	0.3983	0.8514
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.3361	0.5653	0.7389	1.5451
Direct	1	0.2148	0.4710	0.6278	1.6082
Our loop	1	0.2074	0.4694	0.6426	1.6086
Fixed model	10	0.3398	0.5634	0.6944	1.5814
Direct	10	0.3398	0.5634	0.6944	1.5814
Our loop	10	0.3398	0.5634	0.6944	1.5814
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.2556	0.4850	0.6393	1.5874
Direct	1	0.1415	0.4212	0.5835	1.6039
Our loop	1	0.1415	0.4212	0.5835	1.6039
Fixed model	10	0.1548	0.4526	0.6563	1.5806
Direct	10	0.2225	0.5007	0.7096	1.5883
Our loop	10	0.2225	0.5007	0.7096	1.5883
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.1940	0.4500	0.5821	1.6308
Direct	1	0.1542	0.4260	0.5672	1.6190
Our loop	1	0.1542	0.4260	0.5672	1.6190
Fixed model	10	0.2587	0.5020	0.6517	1.6061
Direct	10	0.2239	0.4799	0.6169	1.6075
Our loop	10	0.2239	0.4795	0.6119	1.6075

Table A.5: Paired cluster-bootstrap 95% intervals for each seed and main contrast, in percentage points. Intervals condition on the selected trained models; DTI random-endpoint rows resample drug clusters.

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – fixed (10 labs)	42	+0.33	[-0.06, +0.69]
DTI	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	42	+12.51	[+11.35, +13.71]
DTI	10 – 1 labs (both direct)	42	+5.78	[+5.02, +6.57]
DTI	10 – 1 labs (both loop)	42	+5.78	[+5.02, +6.57]
DTI	Loop – fixed (10 labs)	43	+0.54	[-0.08, +1.11]
DTI	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	43	+7.80	[+6.82, +8.82]
DTI	10 – 1 labs (both direct)	43	+1.65	[+0.90, +2.38]
DTI	10 – 1 labs (both loop)	43	+1.65	[+0.90, +2.38]
DTI	Loop – fixed (10 labs)	44	+5.01	[+4.25, +5.71]
DTI	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	44	+6.60	[+5.61, +7.74]
DTI	10 – 1 labs (both direct)	44	+4.17	[+3.45, +4.93]
DTI	10 – 1 labs (both loop)	44	+4.17	[+3.45, +4.93]
Proteomics	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	42	+13.16	[-0.39, +26.02]
Proteomics	10 – 1 labs (both direct)	42	+16.91	[+3.37, +31.02]
Proteomics	10 – 1 labs (both loop)	42	+16.91	[+3.37, +31.02]
Proteomics	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	43	+15.98	[+1.29, +29.00]
Proteomics	10 – 1 labs (both direct)	43	+17.03	[+3.34, +31.44]
Proteomics	10 – 1 labs (both loop)	43	+17.03	[+3.34, +31.44]
Proteomics	Loop – fixed (10 labs)	44	-0.24	[-3.40, +2.33]
Proteomics	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	44	+6.93	[-3.43, +15.60]
Proteomics	10 – 1 labs (both direct)	44	+16.63	[+3.02, +31.08]
Proteomics	10 – 1 labs (both loop)	44	+16.63	[+3.02, +31.08]
VCC	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	42	+1.70	[-5.36, +8.58]
VCC	10 – 1 labs (both direct)	42	+9.90	[-2.88, +23.04]
VCC	10 – 1 labs (both loop)	42	+9.90	[-2.88, +23.04]
VCC	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	43	+4.83	[-5.50, +16.49]
VCC	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	43	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	43	+15.08	[+2.94, +27.78]
VCC	10 – 1 labs (both loop)	43	+10.24	[-2.74, +23.49]
VCC	Loop – fixed (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	44	-7.44	[-17.23, +1.33]
VCC	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	44	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	44	+6.38	[-4.38, +19.13]
VCC	10 – 1 labs (both loop)	44	+13.82	[+2.54, +26.34]
Norman	Loop – fixed (10 labs)	42	+7.26	[-9.41, +25.41]
Norman	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	42	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	42	+8.59	[-8.30, +26.00]
Norman	10 – 1 labs (both loop)	42	+8.59	[-8.30, +26.00]
Norman	Loop – fixed (10 labs)	43	+6.37	[-11.12, +24.81]

Task	Contrast	Seed	Difference	95% interval
Norman	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	43	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	43	+7.70	[-9.33, +25.85]
Norman	10 – 1 labs (both loop)	43	+7.70	[-9.33, +25.85]
Norman	Loop – fixed (10 labs)	44	+6.67	[-10.37, +24.81]
Norman	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	44	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	44	+8.00	[-8.52, +25.93]
Norman	10 – 1 labs (both loop)	44	+8.00	[-8.52, +25.93]
Tahoe	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	42	+4.48	[-5.97, +14.93]
Tahoe	10 – 1 labs (both direct)	42	+10.45	[-1.49, +22.39]
Tahoe	10 – 1 labs (both loop)	42	+10.45	[-1.49, +22.39]
Tahoe	Loop – fixed (10 labs)	43	-2.99	[-7.46, +0.00]
Tahoe	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	43	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	43	+13.43	[+1.49, +23.88]
Tahoe	10 – 1 labs (both loop)	43	+13.43	[+1.49, +23.88]
Tahoe	Loop – fixed (10 labs)	44	-7.46	[-19.40, +4.48]
Tahoe	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	44	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	44	-2.99	[-17.91, +11.94]
Tahoe	10 – 1 labs (both loop)	44	-2.99	[-17.91, +11.94]

Table A.6: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified six-arm evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Fixed model	10	91.80	89.36
Random	Direct	10	93.76	92.73
Random	Our loop	10	93.76	92.73
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Fixed model	10	90.06	90.40
Unseen drug	Direct	10	91.65	92.58
Unseen drug	Our loop	10	91.65	92.58
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Fixed model	10	85.80	78.92

Endpoint	Method	Labs	AUROC	AP
Unseen protein	Direct	10	87.63	85.07
Unseen protein	Our loop	10	87.63	85.07

Table A.7: Post-hoc TAPB paired AUROC/AP differences and 95% intervals from 1,000 fixed-seed cluster-bootstrap draws. Random and unseen-drug endpoints use drug clusters; unseen-protein uses protein clusters. All three endpoints, six arms and prespecified contrasts are shown.

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Random / 42	Loop — fixed (10 labs)	+0.33	[-0.06, +0.69]	+1.40	[+0.76, +2.05]
Random / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	10 — 1 labs (fixed)	+12.51	[+11.35, +13.71]	+10.56	[+9.10, +12.11]
Random / 42	10 — 1 labs (direct)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	10 — 1 labs (loop)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	Our harness — fixed (1 lab)	+12.84	[+11.82, +14.01]	+11.96	[+10.56, +13.45]
Random / 42	Our harness — direct (1 lab)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Unseen drug / 42	Loop — fixed (10 labs)	+0.10	[-0.47, +0.67]	+1.06	[+0.42, +1.66]
Unseen drug / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	10 — 1 labs (fixed)	+12.30	[+10.77, +13.98]	+7.17	[+5.82, +8.72]
Unseen drug / 42	10 — 1 labs (direct)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	10 — 1 labs (loop)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	Our harness — fixed (1 lab)	+12.40	[+10.94, +13.85]	+8.22	[+6.97, +9.66]
Unseen drug / 42	Our harness — direct (1 lab)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen protein / 42	Loop — fixed (10 labs)	+0.03	[-5.53, +4.28]	+4.12	[-4.10, +8.22]
Unseen protein / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	10 — 1 labs (fixed)	+20.02	[+0.16, +63.83]	+8.14	[-8.27, +51.77]
Unseen protein / 42	10 — 1 labs (direct)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	10 — 1 labs (loop)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	Our harness — fixed (1 lab)	+20.05	[+1.84, +61.27]	+12.25	[-3.01, +53.19]
Unseen protein / 42	Our harness — direct (1 lab)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Random / 43	Loop — fixed (10 labs)	+0.54	[-0.08, +1.11]	+0.44	[-0.54, +1.44]
Random / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	10 — 1 labs (fixed)	+7.80	[+6.82, +8.82]	+7.68	[+6.32, +9.13]
Random / 43	10 — 1 labs (direct)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	10 — 1 labs (loop)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	Our harness — fixed (1 lab)	+8.35	[+7.29, +9.37]	+8.12	[+6.53, +9.76]
Random / 43	Our harness — direct (1 lab)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Unseen drug / 43	Loop — fixed (10 labs)	-0.51	[-1.44, +0.29]	-1.10	[-2.16, -0.18]
Unseen drug / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	10 — 1 labs (fixed)	+6.68	[+5.33, +8.25]	+4.66	[+3.33, +6.36]
Unseen drug / 43	10 — 1 labs (direct)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	10 — 1 labs (loop)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	Our harness — fixed (1 lab)	+6.17	[+4.85, +7.48]	+3.57	[+2.10, +5.09]
Unseen drug / 43	Our harness — direct (1 lab)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen protein / 43	Loop — fixed (10 labs)	-0.96	[-7.24, +3.07]	-1.83	[-21.55, +7.54]
Unseen protein / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	10 — 1 labs (fixed)	+4.79	[-6.61, +28.12]	+0.41	[-16.48, +31.69]
Unseen protein / 43	10 — 1 labs (direct)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	10 — 1 labs (loop)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	Our harness — fixed (1 lab)	+3.83	[-5.83, +22.32]	-1.42	[-14.49, +16.88]
Unseen protein / 43	Our harness — direct (1 lab)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Random / 44	Loop — fixed (10 labs)	+5.01	[+4.25, +5.71]	+8.28	[+7.07, +9.56]
Random / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	10 — 1 labs (fixed)	+6.60	[+5.61, +7.74]	+5.12	[+3.71, +6.80]
Random / 44	10 — 1 labs (direct)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	10 — 1 labs (loop)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	Our harness — fixed (1 lab)	+11.60	[+10.60, +12.65]	+13.40	[+11.88, +15.08]
Random / 44	Our harness — direct (1 lab)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Unseen drug / 44	Loop — fixed (10 labs)	+5.18	[+4.09, +6.19]	+6.57	[+5.28, +7.85]
Unseen drug / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	10 — 1 labs (fixed)	+5.32	[+3.81, +6.97]	+2.04	[+0.38, +3.90]
Unseen drug / 44	10 — 1 labs (direct)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	10 — 1 labs (loop)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Unseen drug / 44	Our harness — fixed (1 lab)	+10.50	[+9.20, +11.87]	+8.61	[+7.10, +10.29]
Unseen drug / 44	Our harness — direct (1 lab)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen protein / 44	Loop — fixed (10 labs)	+6.42	[-2.79, +17.37]	+16.15	[-5.79, +26.26]
Unseen protein / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	10 — 1 labs (fixed)	+3.39	[-16.02, +33.81]	-8.03	[-26.82, +16.99]
Unseen protein / 44	10 — 1 labs (direct)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	10 — 1 labs (loop)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	Our harness — fixed (1 lab)	+9.81	[-2.07, +34.89]	+8.12	[-12.06, +22.71]
Unseen protein / 44	Our harness — direct (1 lab)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]

Three-seed means and sample standard deviations summarize training/search variability on fixed partitions. Sample-level intervals use paired cluster bootstrap with 1,000 fixed-seed resamples of retained predictions. Drug identity defines clusters for the random and unseen-drug interaction sets, and protein sequence defines clusters for the unseen-protein set. Proteomic and cell intervals use compound and intervention clusters, respectively. Both methods in a contrast receive the same re-sampled clusters. The training-allocation study instead uses 5,000 hierarchical paired resamples: training seeds are drawn first, followed by compound or intervention groups within each sampled seed.

The accompanying research records contain the executable configurations, proposals, development histories and saved predictions needed to reconstruct these comparisons. The reproducibility statement describes their availability and verification.

APPENDIX B. LABORATORY, SOURCE AND PROPOSER SENSITIVITY

Sensitivity study design. We examine collaboration scale, source diversity, proposer choice and research budget using the same target scorers as the main study. Client-count experiments cover all five endpoints at $K \in \{1, 2, 5, 10\}$ under both increasing participation and a fixed total data pool. DTI uses seed 61; the lighter proteomics and cell experiments use seeds 61–63. Independent-source DTI and proteomics use seed 61, and external-source VCC uses seeds 61–63. The two-proposer study has two separate protocols: a 12-design, six-slot study on all five endpoints (DTI seed 42; other endpoints seed 61), and a 36-design, 24-slot study on the four lighter endpoints at seed 61. Each candidate receives 100 training rounds. The latter changes the design space as well as the search budget, so budget comparisons are made within a protocol. Main-table results retain their original seeds 42–44 and protocol.

B.1 NUMBER OF LABORATORIES

The participation experiment uses a fixed nested set of the first K original clients, increasing both training and development access. The fixed-pool experiment merges the complete ten-client pool into K training groups while retaining equal-weight evaluation over the original ten development panels. Thus its selection population stays fixed as training partitions change. The identical $K = 10$ fit is reused by both comparisons. Every unique fit uses the default design and 100 rounds. The cell objective forms contrastive negatives within each local training group, so repartitioning changes the negative pool as well as optimization steps. These comparisons characterize the implemented training pipeline at different collaboration scales. Zero between-seed SD can occur when a fixed pretrained head receives an identical single-batch update; it describes training variability on these fixed partitions.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	83.65	84.43	89.62	93.88
PTPC	AP	26.03 ± 5.57	37.38 ± 0.60	29.63 ± 2.50	37.54 ± 0.27
VCC	Macro Top-1	30.02 ± 0.78	39.56 ± 0.11	31.88 ± 0.00	31.48 ± 0.16
Norman	Macro Top-1	25.56 ± 0.00	18.22 ± 0.00	19.68 ± 1.11	15.48 ± 0.00
Tahoe	Macro Top-1	19.40 ± 0.00	25.37 ± 0.00	20.90 ± 0.00	26.37 ± 0.86

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing K also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	85.47	88.96	88.05	93.88
PTPC	AP	34.59 ± 5.66	37.36 ± 1.89	39.19 ± 0.43	37.54 ± 0.27
VCC	Macro Top-1	29.49 ± 0.00	28.80 ± 0.20	31.06 ± 0.22	31.48 ± 0.16
Norman	Macro Top-1	49.63 ± 4.20	19.56 ± 0.20	26.12 ± 0.11	15.48 ± 0.00
Tahoe	Macro Top-1	22.39 ± 0.00	21.39 ± 0.86	20.40 ± 0.86	26.37 ± 0.86

Table B.2: Fixed data pool. All target data are retained and merged into K clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

B.2 INDEPENDENT SOURCES AND TARGET EVALUATION

For DTI and cell transfer, the entire target training pool forms one client, and each auxiliary collection contributes a separate local update through sample-weighted aggregation. Table B.3 distinguishes these source-defined clients from the within-corpus partitions used in the client-count study. Only the original target development panels select the checkpoint; every configuration is evaluated on the

same target test set. Target-only, individual-source and combined-source configurations start from the same task initialization.

Task	Client unit	Sources and biological contexts
DTI	Four data collections	Target BindingDB; BioSNAP; Davis; Human. Each collection contributes one client after pair-level de-duplication.
Cell	Four study/scenario clients	Target VCC; Replogle K562; Nadig HepG2; Jiang IFNB/K562. Each auxiliary source contributes one client.
Proteomics	Ten target partitions plus three source clients	Clients 0–9: PTPC partitions; client 10: Lin (A549, H292, PC9, PSC1); client 11: Ruprecht (A549, Calu6, Calu1, 2030, 2122); client 12: decryptE (Jurkat, five doses).

Table B.3: Source and context mapping in the cross-source experiments. The DTI units are database collections, the cell units preserve their study and biological scenario, and each proteomic auxiliary study owns one client and its response head. The ten PTPC clients remain partitions of one target corpus. Biological contexts within a study are not counted as additional independent studies.

DTI auxiliaries are BioSNAP, Davis and Human from the pinned TAPB release, with 19,092, 1,721 and 2,929 retained training pairs after drug–protein-pair de-duplication against target splits and preceding sources. All five DTI source configurations complete the same 100-round schedule. VCC auxiliary data comprise Replogle K562, Nadig HepG2 and Jiang IFNB/K562, with respectively 60, 60 and 58 retained training interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source feature panels preserve measured genes, and target development/test intervention identities are excluded from source training. The source manifest records retained subsets, feature intersections and exclusions.

Target	Additional source	Metric	Score $\uparrow$	Δ (pp)
DTI	Target only	AUROC	85.47	+0.00
DTI	+ BioSNAP	AUROC	82.44	-3.03
DTI	+ Davis	AUROC	87.86	+2.39
DTI	+ Human	AUROC	87.95	+2.48
DTI	+ All three	AUROC	84.30	-1.18
VCC	Target only	Macro Top-1	29.49 $\pm$ 0.00	+0.00 $\pm$ 0.00
VCC	+ Replogle	Macro Top-1	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08
VCC	+ Nadig	Macro Top-1	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18
VCC	+ Jiang	Macro Top-1	33.07 $\pm$ 0.00	+3.58 $\pm$ 0.00
VCC	+ All three	Macro Top-1	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00
PTPC auxiliary	Target only	AP	34.98	+0.00
PTPC auxiliary	+ Lin	AP	35.01	+0.03
PTPC auxiliary	+ Ruprecht	AP	35.50	+0.52
PTPC auxiliary	+ Both	AP	35.31	+0.34

Table B.4: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean $\pm$ SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. The two-source PTPC protocol shown here uses Lin and Ruprecht; the three-source extension is reported separately in Table B.6. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

The cell study reports mean and SD across three seeds together with a paired intervention-bootstrap analysis. Jiang improves target Top-1 by 3.58 points; its 95% bootstrap interval is $[-2.11, 9.51]$. This positive point estimate motivates further source-aware evaluation, while the complete table also exposes sources whose updates reduce the target metric.

Matched source–scenario clients. Table 2 compares four fixed-design settings: 15 VCC training interventions in one client, all 60 VCC interventions in one client, the same 60 partitioned into four clients, and four source-defined clients with 15 interventions each. The latter uses VCC,

Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562. Both four-client arms share the same 15-intervention VCC anchor. Hash-based sampling fixes the auxiliary identities before fitting, excluding target training, development and test intervention identities as well as duplicates across auxiliary clients. This experiment matches training intervention counts, not raw-cell counts: the same-source arm represents 77,771 cells, versus 33,469 in the cross-scenario arm. Each source retains its measured gene panel and corresponding loss mask.

All four settings use identical head initialization, the default design, 100 rounds and seeds 61–63. The original ten target-development panels select checkpoints; the common target test contains 20 interventions and 360 five-option queries. No proposal search is performed. All twelve fits complete, and saved predictions reproduce the scorer’s metrics under independent evaluation. Cross-scenario Top-1 is lower in each seed, with a mean difference of -1.88 points relative to the same-source four-client setting. The near-zero MRR difference shows that source composition can affect the top-ranked choice differently from the full ranking. The protocol and source identities remain fixed throughout the comparison.

Two-source proteomic transfer and proposal search. Two independent lung-cancer studies provide auxiliary measurements: Lin et al. (Lin et al., 2026) and Ruprecht et al. (Ruprecht et al., 2020). Their continuous response labels differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed source-specific prediction heads. Target fitting uses its binary efficacy objective; Lin predicts continuous CellTiter-Glo response, and Ruprecht predicts an interval-valued potency outcome. Target data retain the original ten clients; Lin and Ruprecht each add one study-level client. Shared encoders use sample-weighted aggregation, whereas each private head is updated only by its own source clients. The common local-worker and coordinator interfaces are reused. This adapter is compared against its own target-only control, separately from the main-table PTPC predictor.

The Lin subset contains 28 drug conditions (20 source-training, eight source-development), spanning five training and two development drugs. Inputs are 24-hour, $10\text{-}\mu\text{M}$ protein log-fold changes, and the response is the mean of the three released CellTiter-Glo ratio measurements. The HCT16/HCT116 alias is excluded from auxiliary training. The Ruprecht subset contains 109 matched drug conditions (85 training, 24 development; 17 and five drugs respectively). Protein log-fold changes at 24 hours are paired with CellTiter-Glo EC50 values in $\log_{10}$ nM: 49 exact, 18 approximate and 42 right-censored outcomes. Squared distance to the encoded potency interval is the loss. The adapter treats explicit lower bounds and values at or above 10,000 nM as right-censored and encodes approximate values with a factor-of-two interval; these are analysis conventions. Profiling dose is not a model input; source acquisition nevertheless uses dose conditions chosen in relation to drug response. The target retains 386 training and 105 development examples, with 148 held-out observations across 92 compounds and 22 positives.

All four source settings receive the same 100-round procedure. A two-proposal Luna loop is also executed independently for each setting. These 12 fits and eight valid proposal calls produce eight fixed/loop evaluations. Fixed and loop selection have identical held-out ranking scores in each source arm (Table B.5). The source effect and the proposal-history effect are thus measured separately.

Training sources	Selection	AP $\uparrow$	Δ vs target only (pp)
Target only	Fixed	34.98	+0.00
Target only	Loop	34.98	+0.00
+ Lin	Fixed	35.01	+0.03
+ Lin	Loop	35.01	+0.03
+ Ruprecht	Fixed	35.50	+0.52
+ Ruprecht	Loop	35.50	+0.52
+ Both	Fixed	35.31	+0.34
+ Both	Loop	35.31	+0.34

Table B.5: Independent-source proteomics auxiliary pilot, seed 61. Fixed design and a two-proposal Luna loop use the same shared-encoder/source-specific-head model and 100-round training allocation. Every source arm and both selection modes are retained. The heldout AP gain from this short loop is zero in all four source conditions; source-transfer gains are a different comparison.

Target-only AP/AUROC are 34.98/74.75. Adding Lin yields 35.01/74.78, Ruprecht 35.50/74.28 and both 35.31/74.17. Paired compound bootstrap (1,000 resamples) gives 95% intervals for the AP changes of $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points, respectively. These results measure transfer from two independent external studies and provide the source-matched controls for the two-proposal comparison.

Three-source proteomic transfer. The third auxiliary study, decryptE (Eckert et al., 2025), measures Jurkat proteomes after 18 hours of drug exposure at 1, 10, 100, 1,000 and 10,000 nM. Its source objective predicts the released continuous metabolic-activity measurement relative to DMSO. We align compounds using the published names, explicit alternative names and structures, and match protein features by UniProt identifier. Inputs comprise log-transformed measured normalized protein intensities, an observation mask, drug fingerprints and dose indicators. Splits group all doses of the same chemical graph together; normalization is fitted on source-training measurements. Target development and test compounds are excluded by canonical identity and stereo-agnostic chemical graph.

The repository’s 145 main processed matrices contain one additional manually classified Vincristine copy; the uniform standard release therefore provides 144 compounds. Pre-training identity checks exclude 12 compounds overlapping the target development/test sets, six records with unsupported internal names and two records with conflicting released chemical-identity/structure annotations. The resulting source contains 124 distinct compounds and 620 dose conditions: 495 training observations from 99 compounds and 125 development observations from 25 compounds. The source manifest records every exclusion and preserves the original annotations.

We append a private continuous-response head to the shared adapter and instantiate this same four-head model in every control. All original encoder and head tensors retain identical seed-61 initialization. The four prespecified arms are target only, target plus decryptE, target plus Lin and Ruprecht, and target plus all three studies. Every arm completes 100 rounds with the same local optimizer and sample-weighted shared-parameter aggregation. Checkpoint selection uses equal-client target-development AP with binary cross-entropy as the tiebreaker; auxiliary development scores are diagnostic. These fixed-design comparisons use zero proposal slots and are separate from the preceding two-source research-loop study.

Additional source	AP $\uparrow$	AUROC $\uparrow$	Δ AP (pp)
Target only	34.98	74.75	+0.00
+ decryptE	35.03	74.21	+0.05
+ Lin + Ruprecht	35.31	74.17	+0.34
+ Lin + Ruprecht + decryptE	35.68	74.28	+0.70

Table B.6: Three-source proteomic transfer on PTPC. All four arms use the same four-head adapter, seed 61, 100-round fitting schedule and target-development checkpoint rule. AP and AUROC are multiplied by 100; bold identifies the best value in each metric. The target test set contains 148 observations from 92 compounds. These are fixed-design source comparisons; the separate two-source proposal study appears in Table B.5.

Combining all three sources raises target AP from 34.98 to 35.68, a 0.70-point gain. Relative to Lin and Ruprecht together, adding decryptE increases AP by 0.37 points and AUROC by 0.11 points. Target-only training retains the highest AUROC across the four arms (74.75 versus 74.28 for all three), showing that the primary AP improvement does not imply uniform gains across ranking metrics. These source-transfer point estimates come from one seed. The four complete CPU fits take 101.21 seconds in total; saved checkpoints and predictions are sealed and independently rescored with exact agreement.

One study per training laboratory. To assign every training client to a distinct source, we additionally merge all 386 target training examples into one client. Lin, Ruprecht and decryptE each remain a separate source client. The four arms use $K = 1, 2, 3, 4$ laboratories and respectively 386, 881, 491 and 986 training conditions. Their four-head architecture, seed-61 initialization, local loss, sample-weighted shared-parameter aggregation and source-private head ownership are unchanged. The merged target worker performs seven minibatches per local epoch, instead of aggregating updates

Table B.7: One worker per proteomic study. The target training pool is merged into one client; Lin, Ruprecht and decryptE each form one additional client. K counts training clients and N counts training conditions. All four fixed-design arms use 100 rounds, the same initialization (seed 61), and equal-weight evaluation over the original ten target development panels. AP and AUROC are percentages on the common target held-out set; best values are bold.

Training sources	K	N	AP $\uparrow$	AUROC $\uparrow$
Target study	1	386	34.41	72.33
+ decryptE	2	881	36.58	73.81
+ Lin + Ruprecht	3	491	36.09	72.69
+ Lin + Ruprecht + decryptE	4	986	37.53	74.39

from ten smaller target partitions. This is therefore a separately trained protocol, with a matching target-only control.

The original ten target-development panels still select checkpoints with equal panel weights, independently of the number of training laboratories. All four configurations complete 100 rounds and use zero proposal slots. Held-out AP and AUROC are independently recomputed from saved predictions, with maximum absolute discrepancy 1.11×10^{-16} . The full CPU run takes 76.18 seconds.

With one client per study, all three auxiliary sources improve target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Adding decryptE to Lin and Ruprecht increases AP by 1.44 points. These single-seed source-transfer comparisons vary available training data as well as participating studies; they measure the effect of those sources under a fixed design, separately from proposal feedback and the matched-condition cell study.

Within-study proteomic contexts. HCC1143, HCC1806 and MDA-MB-453-1 are drawn from the mtPTDS supplementary collection (Sun et al., 2026), each with 47 retained drug conditions and 3,501 proteins shared with the target panel. They are three biological contexts within one study. Target development/test compounds are excluded from source training. These experiments use the original proteomic adapter and seeds 61–63.

Additional context	AP $\uparrow$	Δ AP (pp)	Paired 95% CI
Target only	34.59 ± 5.66	$+0.00 \pm 0.00$	[0.00, 0.00]
+ HCC1143	34.94 ± 5.15	$+0.35 \pm 0.56$	[-1.22, 1.60]
+ HCC1806	35.23 ± 5.35	$+0.64 \pm 0.32$	[-0.78, 1.77]
+ MDA-MB-453-1	35.20 ± 5.31	$+0.61 \pm 0.44$	[-0.85, 2.04]
+ All three	35.47 ± 6.36	$+0.87 \pm 8.89$	[-2.44, 3.99]

Table B.8: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean $\pm$ sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

B.3 RESEARCH-MODEL AND SIX-SLOT PROPOSAL COMPARISON

The short study uses the 12 executable designs of Appendix A. Direct search receives untried identifiers without measured scores; the loop additionally receives accumulated development evidence. Both modes share initialization and the first proposal. Each slot allows two attempts to return the four-field proposal schema; failed slots consume budget and remain in the trace. Every valid candidate restarts from the common task/seed initialization and is fitted for 100 rounds. Prefixes at 0, 1, 2, 4 and 6 slots are evaluated after the trajectory is complete. An exact fit cache can reuse an identical configuration while retaining its logical training-budget charge.

Qwen uses the local sampling settings in Appendix A. The second proposer is the gpt-5.6-luna service, accessed in a text-only session with tool use disabled; request metadata and responses are retained. The comparison holds task messages, editable designs, training procedure and proposal budget fixed across backends. Service-level decoding controls and parameter count are not exposed by the Luna interface.

Task	Proposer	Fixed	Direct@6	Loop@6
DTI	Luna	94.27	94.60	94.27
DTI	Qwen	94.27	94.27	94.27
PTPC	Luna	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27
VCC	Luna	31.50	31.50	31.50
VCC	Qwen	31.50	31.50	31.50
Norman	Luna	15.48	22.30	23.11
Norman	Qwen	15.48	22.30	22.30
Tahoe	Luna	25.37	26.87	26.87
Tahoe	Qwen	25.37	25.37	25.37

Table B.9: Original 12-design/six-slot heldout results. DTI seed 42; four light endpoints seed 61. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
DTI	Luna	94.71	0/0	6/0	6/0	6/6
DTI	Qwen	94.71	0/0	0/0	0/0	6/6
PTPC	Luna	66.72	0/0	0/0	0/0	6/6
PTPC	Qwen	66.72	0/0	0/0	0/0	6/5
VCC	Luna	24.04	3/6	3/6	3/6	6/6
VCC	Qwen	23.65	0/0	0/0	0/0	6/6
Norman	Luna	51.70	5/5	5/5	5/5	6/6
Norman	Qwen	51.70	3/4	3/4	3/4	6/6
Tahoe	Luna	30.00	0/0	0/0	6/3	6/6
Tahoe	Qwen	30.00	0/0	0/0	0/0	4/4

Table B.10: Original 12-design/six-slot search dynamics. DTI seed 42; four light endpoints seed 61. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

All five endpoints have both backends and both search modes. Among the ten direct/loop pairs, Norman/Luna improves by 0.81 Top-1 points, DTI/Luna decreases by 0.33 AUROC points and eight pairs tie. The twenty DTI prefix scores were independently recomputed from saved predictions for all 5,505 test pairs, with checkpoint and output hashes verified. Figure 4 in the main text shows the complete prespecified prefix sequence for every endpoint, alongside its development selection trajectory.

B.4 EXTENDED SEARCH AND TASK-DEPENDENT SATURATION

The extended study increases the design library to 36 configurations: three learning rates, three weight decays, two server-momentum values and two residual-head settings. It runs 24 proposal slots for both backends and modes on PTPC, VCC, Norman and Tahoe. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. The settings for initialization, full candidate fitting, development selection and invalid-slot accounting are otherwise unchanged. DTI’s backend comparison is the six-slot study in B.3; it is not part of this extended protocol.

Task	Proposer	Fixed	Direct@6	Loop@6	Direct@24	Loop@24
PTPC	Luna	37.27	37.27	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27	37.27	37.27
VCC	Luna	31.50	16.94	16.94	20.23	20.01
VCC	Qwen	31.50	31.50	31.50	19.61	19.61
Norman	Luna	15.48	35.48	23.11	23.11	23.11
Norman	Qwen	15.48	22.30	22.30	22.30	22.30
Tahoe	Luna	25.37	13.43	26.87	13.43	13.43
Tahoe	Qwen	25.37	26.87	26.87	13.43	13.43

Table B.11: Extended 36-design/24-slot heldout results. Four light endpoints, seed 61; no native DTI 24-slot result is implied. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
PTPC	Luna	66.72	0/0	0/0	0/0	24/24
PTPC	Qwen	66.72	0/0	0/0	0/0	19/14
VCC	Luna	27.68	16/9	16/10	16/10	24/24
VCC	Qwen	24.83	18/23	18/23	18/23	11/16
Norman	Luna	52.10	22/4	22/4	22/7	24/24
Norman	Qwen	51.70	2/5	2/5	22/19	15/12
Tahoe	Luna	35.00	5/13	5/13	5/13	24/24
Tahoe	Qwen	35.00	18/14	18/14	18/14	14/15

Table B.12: Extended 36-design/24-slot search dynamics. Four light endpoints, seed 61; no native DTI 24-slot result is implied. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

Luna produces valid designs in all 192 attempted slots, and Qwen in 116 of 192. Across 16 mode trajectories, the primary development score is unchanged in the final eight slots in 12 cases; the full score/loss selection key is unchanged in ten. Each trajectory retains 11–24 unvisited designs. A plateau therefore characterizes finite-budget search with a particular backend, rather than exhaustion of the design space.

Table B.12 measures the first slot attaining the smaller terminal development score within each direct/loop pair, a common target reached by both modes. Luna reaches Norman’s common target earlier with feedback (4 versus 22 slots), whereas direct search reaches Tahoe’s target earlier (5 versus 13). Proteomics retains the initial design. Table B.13 connects Norman’s trajectory to concrete configuration changes: reducing the learning rate to 10^{-4} gives an early development gain, and removing the residual head reaches the final primary-score target. Subsequent loss-tiebreak selections can change the retained design without increasing that primary score.

The eight terminal held-out pairs give seven ties and one decrease with feedback (VCC/Luna, -0.22 points). Within this extended protocol, Qwen’s VCC held-out Top-1 decreases from 31.50 at slot 6 to 19.61 at slot 24 while development selection improves. This separates the speed of finding a strong development design from generalization of the selected predictor. The complete development and held-out curves support task-specific budget analysis alongside the proposal-validity and remaining-space diagnostics.

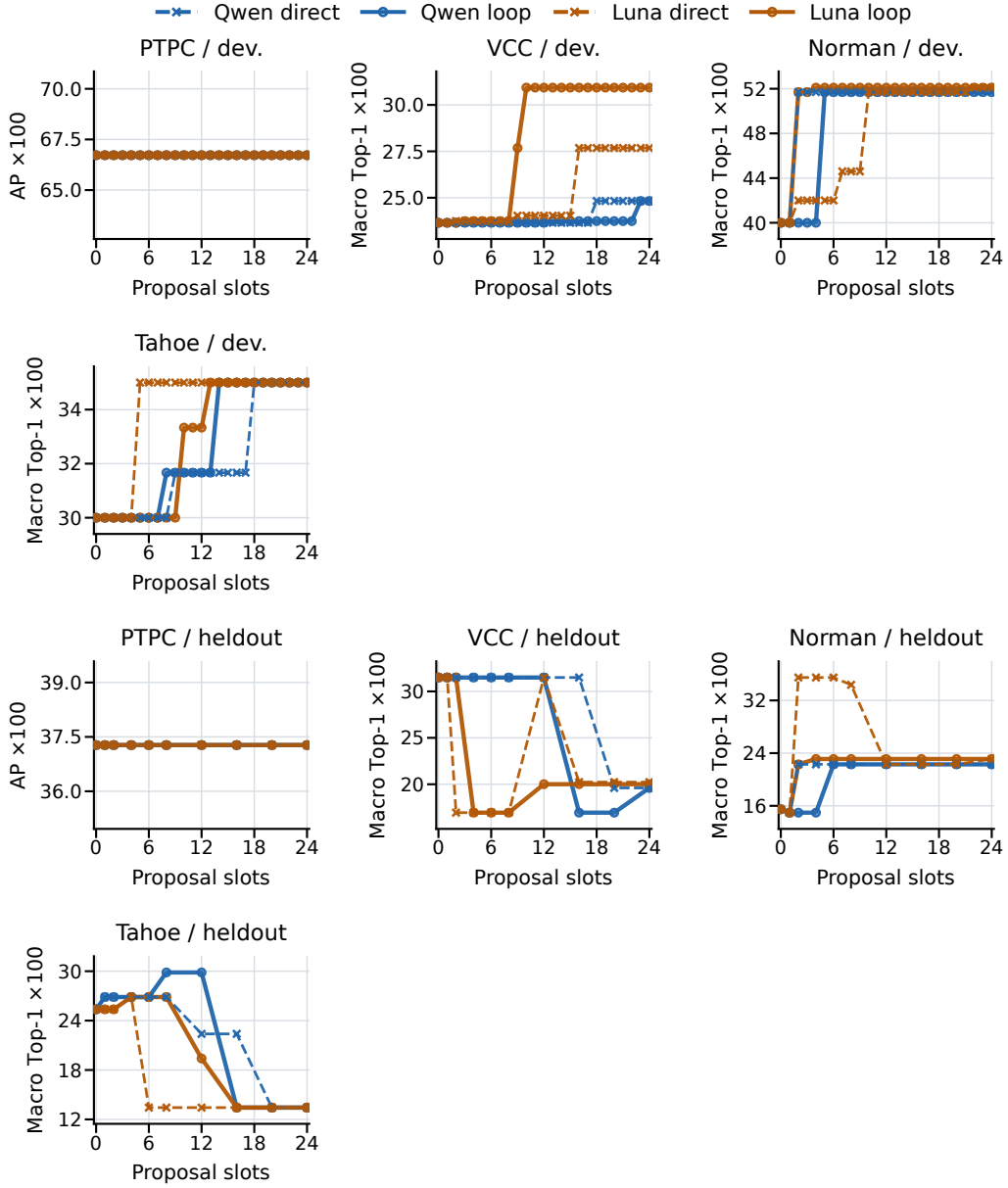


Figure 5: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

B.5 EXECUTED DESIGNS AND PROPOSAL TRACES

Tables B.14 and B.15 decode the two design libraries. The accompanying reproducibility archive retains all 504 attempted proposal slots and their acceptance decisions: 120 slots in the short protocol and 384 in the extended protocol. Its machine-readable proposal trace records each proposal’s full configuration, changes from the incumbent and measured development score; the expanded trace tables are provided with the archive rather than reproduced in the article. Together with the prefix curves, these records connect research decisions to both the executable intervention and its observed outcome.

Mode	Slot	Design	Retained configuration	Dev. Top-1
Direct	0	Initial	Common initial design	40.00
Direct	1	L1W1M0R1	lr=0.001, wd=0.001, m=0, R=1	40.00
Direct	2	L1W1M1R1	lr=0.001, wd=0.001, m=0.9, R=1	42.00
Direct	7	L1W2M1R1	lr=0.001, wd=0.01, m=0.9, R=1	44.60
Direct	10	L0W0M0R1	lr=0.0001, wd=0, m=0, R=1	51.70
Direct	11	L0W1M0R1	lr=0.0001, wd=0.001, m=0, R=1	51.70
Direct	22	L0W1M0R0	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	0	Initial	Common initial design	40.00
Loop	1	L1W1M0R1	lr=0.001, wd=0.001, m=0, R=1	40.00
Loop	2	L0W1M0R1	lr=0.0001, wd=0.001, m=0, R=1	51.70
Loop	4	L0W1M0R0	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	7	L0W0M0R0	lr=0.0001, wd=0, m=0, R=0	52.10

Table B.13: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

Design	Learning rate	Weight decay	Server momentum	Residual head
D00	0.001	0.001	0	No
D01	0.0001	0.001	0	No
D02	0.01	0.001	0	No
D03	0.001	0	0	No
D04	0.001	0.01	0	No
D05	0.001	0.001	0.9	No
D06	0.001	0.001	0	Yes
D07	0.001	0.001	0.9	Yes
D08	0.0001	0.001	0	Yes
D09	0.0001	0.001	0.9	Yes
D10	0.01	0.01	0	No
D11	0.001	0.01	0	Yes

Table B.14: Twelve executable configurations for the short6 protocol. Every design has proximal coefficient zero. A trace design ID uniquely identifies these parameter changes; the task-specific prediction interface remains fixed.

Design	Learning rate	Weight decay	Server momentum	Residual head
L0W0M0R0	0.0001	0	0	No
L0W0M0R1	0.0001	0	0	Yes
L0W0M1R0	0.0001	0	0.9	No
L0W0M1R1	0.0001	0	0.9	Yes
L0W1M0R0	0.0001	0.001	0	No
L0W1M0R1	0.0001	0.001	0	Yes
L0W1M1R0	0.0001	0.001	0.9	No
L0W1M1R1	0.0001	0.001	0.9	Yes
L0W2M0R0	0.0001	0.01	0	No
L0W2M0R1	0.0001	0.01	0	Yes
L0W2M1R0	0.0001	0.01	0.9	No
L0W2M1R1	0.0001	0.01	0.9	Yes
L1W0M0R0	0.001	0	0	No
L1W0M0R1	0.001	0	0	Yes
L1W0M1R0	0.001	0	0.9	No
L1W0M1R1	0.001	0	0.9	Yes
L1W1M0R0	0.001	0.001	0	No
L1W1M0R1	0.001	0.001	0	Yes
L1W1M1R0	0.001	0.001	0.9	No
L1W1M1R1	0.001	0.001	0.9	Yes
L1W2M0R0	0.001	0.01	0	No
L1W2M0R1	0.001	0.01	0	Yes
L1W2M1R0	0.001	0.01	0.9	No
L1W2M1R1	0.001	0.01	0.9	Yes
L2W0M0R0	0.01	0	0	No
L2W0M0R1	0.01	0	0	Yes
L2W0M1R0	0.01	0	0.9	No
L2W0M1R1	0.01	0	0.9	Yes
L2W1M0R0	0.01	0.001	0	No
L2W1M0R1	0.01	0.001	0	Yes
L2W1M1R0	0.01	0.001	0.9	No
L2W1M1R1	0.01	0.001	0.9	Yes
L2W2M0R0	0.01	0.01	0	No
L2W2M0R1	0.01	0.01	0	Yes
L2W2M1R0	0.01	0.01	0.9	No
L2W2M1R1	0.01	0.01	0.9	Yes

Table B.15: Thirty-six executable configurations for the long24 protocol. Every design has proximal coefficient zero. A trace design ID uniquely identifies these parameter changes; the task-specific prediction interface remains fixed.